

Philosophy of Science

“Philosophy of science is the branch of philosophy that explore the nature of scientific inquiry including the scientific methods, the nature of scientific knowledge and the relationship between science and philosophy.”

Basically, philosophy of science provides a deeper understanding of the scientific enterprise and its role in human society.

i What is Science:-

“Science is the systematic and structured way of acquiring knowledge about the world through observation and Experiment.”

Science involves:

- ♦ Asking questions
- ♦ Making Observations
- ♦ Testing ideas
- ♦ Finding answers

→ Science help us to understand the world around us, the things around us, how these things work and why things happen.

Branches :-

There are so many branches of science and some of them are given below;

Natural science :-

“It's deals with the study of natural world.”

e.g:- Biology, chemistry and Physics.

Formal Science :-

“It's deal with the study of logic, mathematic's and statistic's.”

Social Science :-

“It's deal with the study of human behavior and societies.”

e.g :-

Psychology, sociology, anthropology.

Applied Sciences :-

“It's deal with the study of practical application of scientific knowledge.”

e.g :-

Engineerings, medicine etc.

What is Scientific method :-

“Scientific method is a systematic process used to develop and test scientific knowledge.”

Scientific method involves following steps:-

1. Observation:-

Make observation about the world around you.

2. Question:-

Ask a questions about what you have observed.

3. Hypothesis:-

Propose a possible explanation for what you have observed.

4. Prediction:-

Make predictions based on your hypothesis.

5. Experimentation:-

Design and conduct experiments to test your hypothesis.

6. Data analysis:-

Analyze the data collected during experimentation.

Conclusion :-

Draw conclusions based on your analysis.

Replication :-

Repeat the experim to verify the results.

iii. The problem of Induction:-

“The Problem of induction is a philosophical problem that questions the rationality of predictions about unobserved things based on previous observations.”

Method

→ We make observation and gather data.

- We use these data to make general rules or predictions.
- But there is no guarantee that those rules will always hold true in the future.

Examples:-

- ◆ Just because the sun has risen every morning so far, does not mean it will definitely rise tomorrow.
- ◆ You eat 10 apples and none of them are rotten. You assume all apples are good but the next one apple you eat might be rotten.

These two examples illustrate how induction can lead to incorrect assumptions. Just because something

happens in a certain way multiple times,
does not mean it's will always happen
that way.

Karl Popper's theory of Scientific Falsificationism

“Karl Popper's falsificationism is
a philosophical approach to scientific
inquiry that emphasizes the role
of criticism and falsification
in the development of scientific
knowledge.”

Method

→ A general statement and theory
that claims to be true for all
cases.

→ The Process of testing the claim through

Experiments, observation and data analysis.

→ If the test find the contradictory evidence, the claim is prove wrong. But if the test fail to find the contradictory evidence, the claim is accepted as true, but remains open to future falsification.

Example 2:-

Claim: Some one says, "All swans are white"

Test: you go to lake and observe swans. If you see a black swan, the claim is falsified.

Conclusion: claim of "all swans are white" is proven wrong.

In this example, the

claim is tested and proven wrong by a single counter example. This illustrates popper's idea that scientific theories can be falsified by contradictory evidence.

Thomas Kuhn's theory of Scientific Revolutions :-

“According to Thomas Kuhn's theory, scientific revolution is a transformative event in which a new Paradigm replaces an existing one.”

Example :-

Einsteinian Revolution : The replacement of classical mechanics with relativity theory.

key point :-

Crisis :-

A scientific revolution often follows a period of crisis, during which the existing Paradigm is challenged by contradictions.

- ◆ New Paradigm:-

The emergence of new paradigm that provides a comprehensive explanation of phenomenon in question.

- ◆ Paradigm shift:-

The transition from the old paradigm to new one, which involves a fundamental change in way scientists think about and approach the field.
